

## Procurement and Sourcing

### What is Procurement?

Procurement is a process of maintaining a long term relationship with the customer and supplier, there can be price negotiations between them.

1<sup>st</sup> to identify what is the customer need, what product they are looking for.

2<sup>nd</sup> research to get the best fit supplier.

3<sup>rd</sup> request for proposal.

4<sup>th</sup> agreeing on pricing and delivery terms and SLA

5<sup>th</sup> purchasing the actual order from the supplier.

6<sup>th</sup> verification on the quality and quantity of the products received.

7<sup>th</sup> Approve vendors invoice and process final payment.

**This is not a one-time buying and selling process.**

| Stakeholder                        | Responsibilities                                                                                         |
|------------------------------------|----------------------------------------------------------------------------------------------------------|
| <b>Customer (Buyer Company)</b>    | Creates material requirements, tracks procurement requests, monitors order status.                       |
| <b>Procurement Executive</b>       | Creates RFQs, manages suppliers, negotiates quotations, generates Purchase Orders.                       |
| <b>Procurement Manager</b>         | Approves requirements, RFQs, supplier selection, contracts, SLAs, and Purchase Orders.                   |
| <b>Supplier (Vendor)</b>           | Registers on the platform, responds to RFQs, accepts Purchase Orders, uploads invoices, tracks payments. |
| <b>Material Planning Executive</b> | Reviews material demand, checks shortages, creates purchase requisitions.                                |
| <b>Inventory Planner</b>           | Monitors stock availability, reserves materials, allocates inventory, plans internal transfers.          |
| <b>Finance Executive</b>           | Verifies invoices, performs invoice matching, processes payments.                                        |
| <b>Finance Manager</b>             | Approves payments, reviews penalties, monitors financial reports.                                        |
| <b>Auditor</b>                     | Reviews procurement activities, financial transactions, compliance, and audit logs.                      |

| Stakeholder                 | Responsibilities                                                     |
|-----------------------------|----------------------------------------------------------------------|
| <b>System Administrator</b> | Manages users, roles, permissions, system settings, and master data. |

# EPIC 1 – Platform Administration

## Module 1 – Authentication & User Management

| PBI ID | User Story                                                                    | Tasks                                                             | Priority |
|--------|-------------------------------------------------------------------------------|-------------------------------------------------------------------|----------|
| PB-001 | As a Customer/Supplier/Employee, I want to securely log into the system.      | Login UI, Authentication API, Forgot Password, Session Management | High     |
| PB-002 | As a Customer, I want to register my organization.                            | Customer Registration Form, Validation, Company Creation          | High     |
| PB-003 | As a Supplier, I want to register my company.                                 | Supplier Registration, Document Upload, Validation                | High     |
| PB-004 | As an Admin, I want to approve or reject customer and supplier registrations. | Approval Dashboard, Status Update, Notification                   | High     |

This module provides secure access to the procurement platform by enabling authentication for customers, suppliers, employees, and administrators. It includes user registration, login, password recovery, session management, and administrator approval workflows to ensure that only verified organizations and authorized users can access the system.

## Module 2 – Employee & Role Management

| PBI ID | User Story                                           | Tasks                                                        | Priority |
|--------|------------------------------------------------------|--------------------------------------------------------------|----------|
| PB-005 | As an Admin, I want to create employee accounts.     | Employee Creation Form, Temporary Password Generation        | High     |
| PB-006 | As an Admin, I want to assign departments and roles. | Department Assignment, Role Assignment, Permission Mapping   | High     |
| PB-007 | As an Admin, I want to manage users.                 | Activate/Deactivate Users, Reset Password, Edit User Details | Medium   |
| PB-008 | As an Admin, I want to configure role permissions.   | RBAC Configuration, Permission Matrix                        | Medium   |

This module enables administrators to manage internal users by creating employee accounts, assigning departments and roles, configuring permissions, and maintaining role-based access control (RBAC). It ensures users have appropriate access privileges based on their responsibilities.

## EPIC 2 – Procurement & Sourcing

### Module 3 – Requirement Management

| PBI ID | User Story                                 | Tasks                                     | Priority |
|--------|--------------------------------------------|-------------------------------------------|----------|
| PB-009 | Customer submits a material requirement.   | Requirement Form, File Upload, Validation | High     |
| PB-010 | Procurement reviews customer requirements. | Review Screen, Approval Workflow, Remarks | High     |
| PB-011 | Customer tracks requirement progress.      | Status Timeline, Search, Notifications    | Medium   |
| PB-012 | View requirement history.                  | Audit Trail, Requirement Archive          | Medium   |

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This module allows customers to submit procurement requirements and enables procurement teams to review, approve, and monitor those requests. It also provides requirement tracking, notifications, and historical records for complete visibility throughout the procurement lifecycle.

### Module 4 – Supplier Management

| PBI ID | User Story                         | Tasks                                     | Priority |
|--------|------------------------------------|-------------------------------------------|----------|
| PB-013 | Maintain supplier profiles.        | Supplier CRUD, Product Categories         | High     |
| PB-014 | Verify supplier documents.         | Certificate Verification, Expiry Tracking | High     |
| PB-015 | Evaluate supplier performance.     | Rating, Delivery History, Quality History | Medium   |
| PB-016 | Blacklist or reactivate suppliers. | Status Management, Restrict Procurement   | Medium   |

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This module manages supplier information throughout their lifecycle, including registration, document verification, performance evaluation, and supplier status management. It helps organizations maintain reliable supplier relationships while ensuring compliance with procurement standards.

## Module 5 – RFQ & Quotation

| PBI ID | User Story                              | Tasks                                   | Priority |
|--------|-----------------------------------------|-----------------------------------------|----------|
| PB-017 | Create RFQ.                             | RFQ Form, Product Selection             | High     |
| PB-018 | Send RFQ to suppliers.                  | Supplier Selection, RFQ Distribution    | High     |
| PB-019 | Suppliers submit quotations.            | Quotation Submission, Attachment Upload | High     |
| PB-020 | Compare quotations and select supplier. | Comparison Screen, Supplier Selection   | High     |

This module supports the Request for Quotation (RFQ) process by allowing procurement teams to create RFQs, distribute them to suppliers, receive quotations, and compare supplier responses. It simplifies supplier selection by providing structured quotation evaluation and comparison.

## Module 6 – Contract & SLA

| PBI ID | User Story                 | Tasks                                        | Priority |
|--------|----------------------------|----------------------------------------------|----------|
| PB-021 | Create contracts.          | Contract Form, Document Repository           | High     |
| PB-022 | Configure SLA.             | Delivery Terms, Quality Rules, Penalty Rules | High     |
| PB-023 | Manage contract lifecycle. | Renewal, Version History, Expiry Alerts      | Medium   |
| PB-024 | View contracts.            | Contract Dashboard, Search, Filters          | Medium   |

This module manages procurement contracts and Service Level Agreements (SLAs). It enables contract creation, SLA configuration, lifecycle management, renewal tracking, and document storage while ensuring suppliers comply with agreed delivery, quality, and penalty terms.

## Module 7 – Purchase Order

| PBI ID | User Story                                   | Tasks                                       | Priority |
|--------|----------------------------------------------|---------------------------------------------|----------|
| PB-025 | Generate Purchase Orders.                    | PO Creation, Item Management                | High     |
| PB-026 | Approve Purchase Orders.                     | Approval Workflow, Notifications            | High     |
| PB-027 | Supplier accepts or rejects Purchase Orders. | PO Response Portal                          | High     |
| PB-028 | Track Purchase Orders.                       | Delivery Timeline, Amendments, Cancellation | Medium   |

This module facilitates the creation, approval, and management of Purchase Orders (POs). Suppliers can respond to purchase orders, while procurement teams can monitor order status, amendments, cancellations, and delivery progress from a centralized dashboard.

# EPIC 3 – Resource Planning

## Module 8 – Material Requirement Planning

| PBI ID | User Story                       | Tasks                                       | Priority |
|--------|----------------------------------|---------------------------------------------|----------|
| PB-029 | Maintain Bill of Materials.      | BOM CRUD, Version Management                | High     |
| PB-030 | Calculate material requirements. | Requirement Calculation, Shortage Detection | High     |
| PB-031 | Generate procurement plan.       | Procurement Quantity Calculation            | High     |
| PB-032 | View planning dashboard.         | Planning Reports, Export                    | Medium   |

This module performs material planning by maintaining Bills of Materials (BOM), calculating required quantities, identifying shortages, and generating procurement plans. It ensures sufficient materials are available to meet production and customer demand efficiently.

## Module 9 – Purchase Requisition

| PBI ID | User Story                    | Tasks                              | Priority |
|--------|-------------------------------|------------------------------------|----------|
| PB-033 | Raise Purchase Requisition.   | PR Form, Validation                | High     |
| PB-034 | Raise emergency requisition.  | Emergency Workflow, Priority Rules | High     |
| PB-035 | Approve Purchase Requisition. | Approval Workflow                  | High     |
| PB-036 | Track Purchase Requisition.   | Dashboard, Search, Notifications   | Medium   |

This module enables departments to raise standard and emergency purchase requisitions for required materials. It supports approval workflows, requisition tracking, and notifications, ensuring procurement requests are processed efficiently and transparently.

## Module 10 – Stock Visibility

| PBI ID | User Story                         | Tasks                              | Priority |
|--------|------------------------------------|------------------------------------|----------|
| PB-037 | Monitor stock across warehouses.   | Warehouse Dashboard, Stock Summary | High     |
| PB-038 | View available and reserved stock. | Reservation Logic, Available Stock | High     |
| PB-039 | Monitor low stock.                 | Safety Stock, Alerts               | High     |
| PB-040 | Generate inventory reports.        | Reports, Filters, Export           | Medium   |

This module provides real-time visibility into inventory across multiple warehouses. It enables users to monitor available and reserved stock, identify low-stock situations, generate inventory reports, and support effective inventory planning and replenishment.

## Module 11 – Material Reservation & Transfer

| PBI ID | User Story                         | Tasks                                | Priority |
|--------|------------------------------------|--------------------------------------|----------|
| PB-041 | Reserve materials for departments. | Reservation Workflow, Approval       | High     |
| PB-042 | Allocate materials.                | Allocation Logic, Department Mapping | High     |
| PB-043 | Request internal transfers.        | Transfer Request, Approval           | High     |
| PB-044 | Track reservations and transfers.  | Dashboards, Reports                  | Medium   |

This module manages internal material reservations and warehouse transfers. Departments can reserve inventory, allocate available materials, request inter-department transfers, and monitor reservation and transfer status through dashboards and reports to ensure optimal resource utilization.

## EPIC 4 – Finance & Audit

### Module 12 – Invoice Verification

| PBI ID | User Story                          | Tasks                                 | Priority |
|--------|-------------------------------------|---------------------------------------|----------|
| PB-045 | Upload supplier invoices.           | Invoice Upload, Validation            | High     |
| PB-046 | Perform three-way matching.         | Invoice ↔ PO ↔ Goods Receipt Matching | High     |
| PB-047 | Validate taxes and invoice details. | GST Validation, Invoice Verification  | Medium   |
| PB-048 | Approve or reject invoices.         | Approval Workflow, Notifications      | High     |

This module streamlines supplier invoice processing by supporting invoice uploads, performing three-way matching between invoices, purchase orders, and goods receipts, validating tax information, and managing invoice approval workflows before payment processing.

### Module 13 – Payment & SLA Compliance

| PBI ID | User Story                       | Tasks                               | Priority |
|--------|----------------------------------|-------------------------------------|----------|
| PB-049 | Schedule and process payments.   | Payment Workflow, Vendor Ledger     | High     |
| PB-050 | Monitor supplier SLA compliance. | Delivery Tracking, Quality Tracking | High     |
| PB-051 | Calculate penalties.             | Penalty Engine, Exception Handling  | High     |
| PB-052 | Maintain payment history.        | Reports, Dashboard                  | Medium   |

This module manages vendor payments while ensuring supplier compliance with contractual Service Level Agreements. It schedules payments, tracks supplier delivery and quality

performance, calculates penalties for SLA violations, and maintains comprehensive payment records and financial dashboards.

## Module 14 – Audit & Analytics

| PBI ID | User Story                                 | Tasks                                   | Priority |
|--------|--------------------------------------------|-----------------------------------------|----------|
| PB-053 | Maintain audit logs.                       | User Activity Logging, Transaction Logs | High     |
| PB-054 | Generate procurement analytics.            | Procurement KPIs, Cost Analysis         | High     |
| PB-055 | Generate supplier performance reports.     | Delivery Performance, Quality Metrics   | Medium   |
| PB-056 | Generate financial and compliance reports. | Financial Reports, Export, Dashboards   | Medium   |

This module provides auditing and business intelligence capabilities for the procurement platform. It maintains detailed audit logs, generates procurement and supplier performance analytics, produces financial and compliance reports, and offers dashboards to support informed decision-making and regulatory compliance.

## What will make it successful

### 1. One common database

Do **not** let every member create their own database.

Instead, design one shared database.

Example:

- Users
- Organizations
- Suppliers
- Employees
- Departments
- Roles
- Requirements
- RFQs
- Quotations
- Contracts
- SLAs
- Purchase Orders

- Goods Receipts
- Inventory
- Reservations
- Transfers
- Invoices
- Payments
- Audit Logs

Everyone works on the same schema.

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## 2. One backend

Avoid 14 separate backend servers.

Instead:

Backend

Authentication Module  
Supplier Module  
RFQ Module  
Purchase Module  
Inventory Module  
Finance Module

↓

Single Express Server

↓

Single PostgreSQL Database

Each member owns one folder.

Example

backend/

routes/

auth/  
employee/  
supplier/  
rfq/  
contract/  
purchaseorder/  
inventory/  
invoice/  
payment/  
audit/

controllers/



services/

models/

---

## 3. One frontend

Instead of

14 React projects

Create

One React Project

src/

pages/

Authentication/

Employees/

Requirements/

Suppliers/

RFQ/

Contracts/

PurchaseOrders/

MRP/

Inventory/

Finance/

Audit/

Each member develops one folder.

---

## Integration Flow

Your modules already connect nicely.

Authentication

↓

Role Management

↓

Requirement Management

↓

Supplier Management

↓

RFQ

↓

Quotation

↓

Contract

↓

Purchase Order

↓

Material Planning

↓

Purchase Requisition

↓

Inventory

↓

Material Reservation

↓

Invoice Verification

↓

Payment

↓

Audit

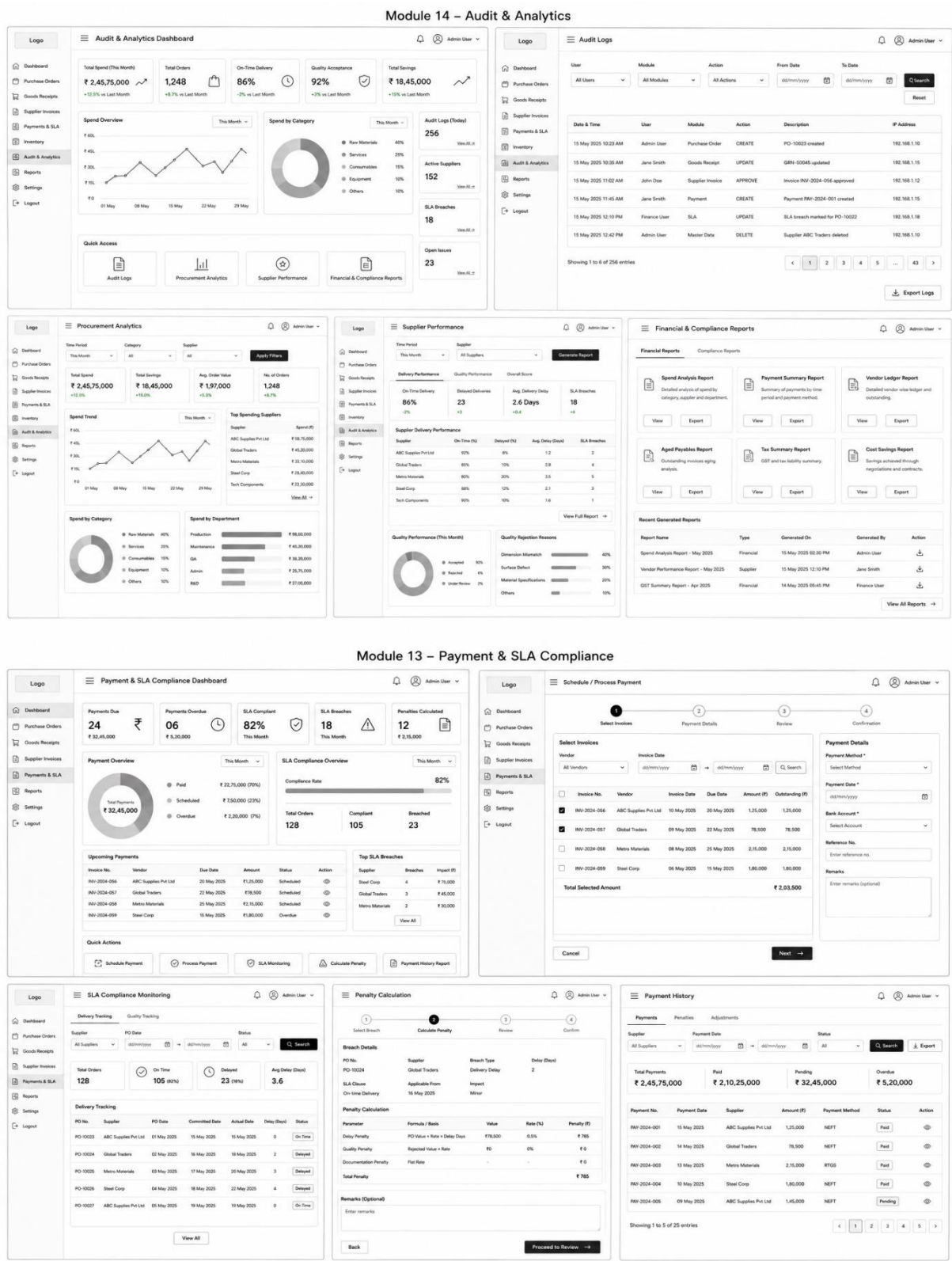
That is a complete procurement lifecycle.

## Common UI

Use

- Same sidebar
- Same navbar
- Same buttons
- Same color theme
- Same fonts
- Same cards
- Same tables

Otherwise it will look like 14 different projects.



Module 12 – Supplier Invoice Management

Logo

Supplier Invoice Dashboard

Total Invoices

128

Pending Matching

28

Pending Approval

18

Approved Invoices

64

Rejected Invoices

18

Invoice Status Overview

Pending Upload

12

Pending Matching

28

Pending Validation

20

Pending Approval

18

Approved

64

Rejected

18

Recent Invoices

| Invoice No.  | Supplier             | PO No.   | Invoice Date | Amount    | Status            |
|--------------|----------------------|----------|--------------|-----------|-------------------|
| INV-2024-056 | ABC Supplies Pvt Ltd | PO-10023 | 10 May 2025  | ₹1,25,000 | Pending Matching  |
| INV-2024-056 | Global Traders       | PO-10022 | 09 May 2025  | ₹18,500   | Pending Invoicing |
| INV-2024-054 | Metals Materials     | PO-10021 | 08 May 2025  | ₹2,15,000 | Pending Approval  |
| INV-2024-053 | ABC Supplies Pvt Ltd | PO-10020 | 07 May 2025  | ₹95,000   | Approved          |
| INV-2024-052 | Steel Corp           | PO-10019 | 06 May 2025  | ₹1,80,000 | Rejected          |

Quick Actions

Upload Invoice

Three Way Matching

Invoice Validation

View Reports

Upload Supplier Invoice

1

2

3

4

Invoice Details

Supplier \*

Select Supplier

PO Number \*

Select PO Number

Invoice Number \*

Enter Invoice Number

Invoice Date \*

dd/mm/yyyy

Invoice Amount \*

₹

Error Amount

Description (Optional)

Enter description

Upload Invoice Document \*

Drag & Drop file here

Browse File

Supported formats: PDF, JPG, PNG (Max 10MB)

Uploaded File

No file uploaded yet.

Three-Way Matching

1

2

3

Invoice

Supplier

PO No.

Invoice Amount

Invoice Date

Invoice No.

INV-2024-056

Supplier

ABC Supplies Pvt Ltd

PO No.

PO-10023

Invoice Amount

₹1,25,000

Invoice Date

10 May 2025

Match

Invoice

Purchase Order

Goods Receipt

Match Summary

| Category | Invoice   | PO        | GRN       | Match Status | Remarks                |
|----------|-----------|-----------|-----------|--------------|------------------------|
| Item     | 10        | 10        | 10        | Matched ✓    | All items matched      |
| Quantity | 100       | 100       | 100       | Matched ✓    | All quantities matched |
| Rate     | ₹1,25,000 | ₹1,25,000 | ₹1,25,000 | Matched ✓    | All rates matched      |
| Amount   | ₹1,25,000 | ₹1,25,000 | ₹1,25,000 | Matched ✓    | Amounts are equal      |

Invoice Validation

1

2

3

Supplier Invoice

Supplier

PO No.

Invoice Amount

Invoice Date

Invoice No.

INV-2024-056

Supplier

ABC Supplies Pvt Ltd

PO No.

PO-10023

Invoice Amount

₹1,25,000

Invoice Date

10 May 2025

Validation Check

| Check                 | Details         | Status | Remarks                   |
|-----------------------|-----------------|--------|---------------------------|
| Validation Check      | 33AMCOUN000021  | Valid  | GSTIN is valid            |
| Invoice Number Format | INV-2024-056    | Valid  | Format is correct         |
| Invoice Date          | 10 May 2025     | Valid  | Invoice date is valid     |
| GST Calculation       | ₹18,047 (18%)   | Valid  | Tax calculated correctly  |
| Tax Amount            | ₹18,047         | Valid  | Tax amount is correct     |
| Total Amount          | ₹1,25,000       | Valid  | Total matches calculation |
| Place of Supply       | Tamil Nadu (IN) | Valid  | Place of supply valid     |

Tax Summary

| Supplier Invoice | GST (IN) | GST (IN) | GST (IN) | Total Tax | Total Amount |
|------------------|----------|----------|----------|-----------|--------------|
| ₹1,25,000        | ₹18,046  | ₹18,046  | ₹0       | ₹36,092   | ₹1,25,000    |

Invoice Approval

1

2

3

4

Review Summary

Three-Way Match

Valid

No Validation

Valid

Invoice Amount

₹1,25,000

Overall Status

Ready for Approval

Common Comments

Enter comments

Attachments

No additional documents

+ Add Attachment

Approval Action

Approve

Reject

Return for Changes

Approval History

| Action By       | Role | Action | Comments | Date & Time |
|-----------------|------|--------|----------|-------------|
| No entries yet. |      |        |          |             |

Notifications

Send notifications to:

Supplier Contact

Finance Team

Procurement Team

Module 11 – Material Reservation & Transfer

Material Reservation & Transfer Dashboard

Active Reservations

18

Pending Approvals

06

In-Transit Transfers

08

Completed (This Month)

24

Recent Reservations

| Reservation ID | Department  | Material       | Qty | Status   | Date        |
|----------------|-------------|----------------|-----|----------|-------------|
| RES-00018      | Production  | M5 Plate 10mm  | 50  | Approved | 01 May 2025 |
| RES-00017      | Maintenance | Bearing 6205   | 20  | Pending  | 30 Apr 2025 |
| RES-00016      | QA          | Measuring Tape | 10  | Approved | 28 Apr 2025 |
| RES-00015      | R&D         | Aluminium Rod  | 15  | Rejected | 28 Apr 2025 |

Recent Transfers

| Transfer ID | From Dept. | To Dept.    | Material        | Status     |
|-------------|------------|-------------|-----------------|------------|
| TRF-00023   | Stores     | Production  | M5 Plate 10mm   | In Transit |
| TRF-00022   | Production | Maintenance | Bearing 6205    | Approved   |
| TRF-00021   | Stores     | QA          | Vernier Caliper | Pending    |
| TRF-00020   | R&D        | Production  | Aluminium Rod   | Completed  |

Quick Actions

Reserve Materials

Request Transfer

Allocate Materials

View Reports

Reserve Materials

1

2

3

Reservation Details

Reservation ID

Auto Generated

Requesting Department \*

Select Department

Required Date \*

dd/mm/yyyy

Material \*

Select Material

Purpose / Remarks

Enter purpose or remarks

Requested Items

No items added yet.

+ Add Item

Request Internal Transfer

1

2

3

Transfer Information

Transfer ID

Auto Generated

From Department \*

Select Department

To Department \*

Select Department

Required Date \*

dd/mm/yyyy

Material Items

| # | Material *      | Available Qty | Quantity * | UOM | Action |
|---|-----------------|---------------|------------|-----|--------|
| 1 | Select Material |               | Enter Qty  |     |        |

+ Add Another Item

Allocate Materials

Reservation ID

RES-00018

Department

Production

Required Date

01 May 2025

Items to Allocate

| # | Material      | Requested Qty | Available Qty | Allocate Qty | UOM |
|---|---------------|---------------|---------------|--------------|-----|
| 1 | M5 Plate 10mm | 50            | 120           | 50           | Nos |
| 2 | Hex Bolt M10  | 100           | 250           | 100          | Nos |
| 3 | Washer 10mm   | 200           | 500           | 200          | Nos |

Allocation Details

Allocate From Location / Bin \*

Select Location / Bin

Reference Document (Optional)

Choose File

No file chosen

Remarks

Enter remarks (optional)

Reservations & Transfers

Reservations

Transfers

Status

All

Department

All

From Date

dd/mm/yyyy

To Date

dd/mm/yyyy

Search

Search

Export

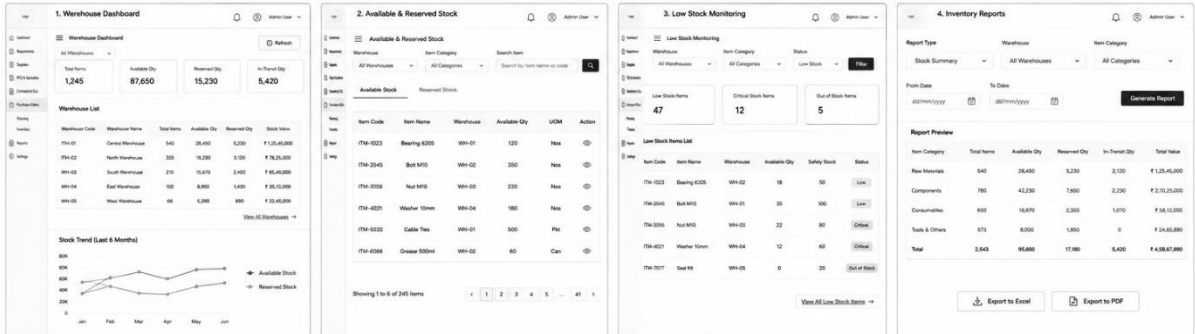
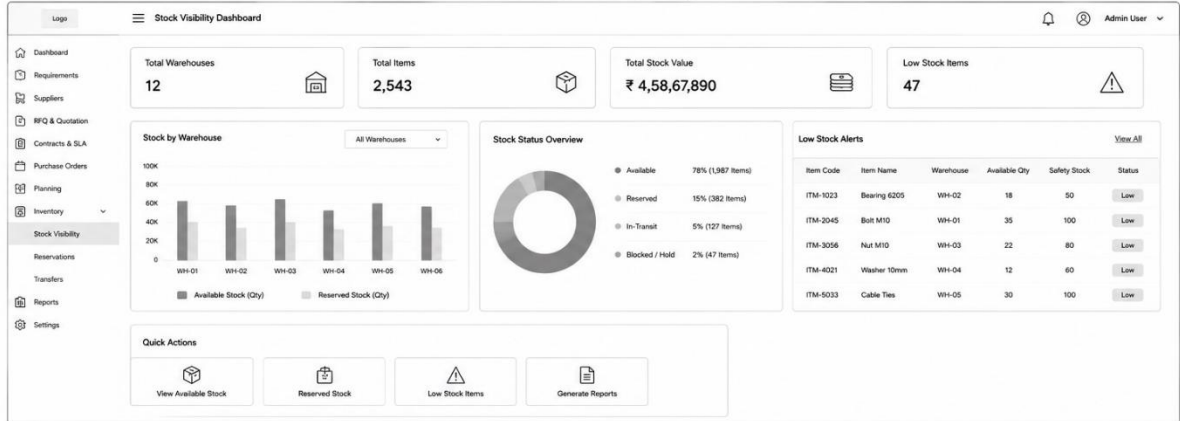
Export

Table

| ID        | Type        | From Dept. | To Dept.    | Material        | Qty | Status     | Date        | Action |
|-----------|-------------|------------|-------------|-----------------|-----|------------|-------------|--------|
| RES-00018 | Reservation | -          | Production  | M5 Plate 10mm   | 50  | Approved   | 01 May 2025 |        |
| RES-00017 | Reservation | -          | Maintenance | Bearing 6205    | 20  | Pending    | 30 Apr 2025 |        |
| TRF-00023 | Transfer    | Stores     | Production  | M5 Plate 10mm   | 50  | In Transit | 01 May 2025 |        |
| TRF-00022 | Transfer    | Production | Maintenance | Bearing 6205    | 20  | Approved   | 30 Apr 2025 |        |
| TRF-00021 | Transfer    | Stores     | QA          | Vernier Caliper | 05  | Pending    | 29 Apr 2025 |        |

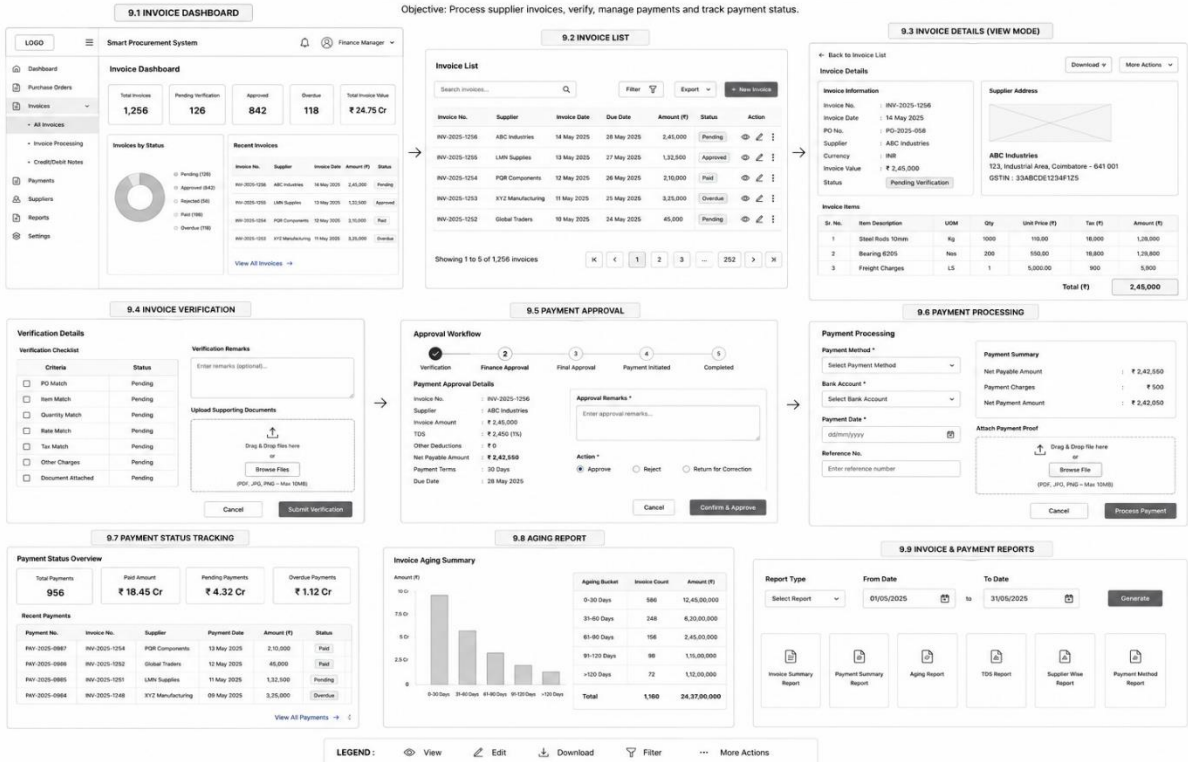
Showing 1 to 5 of 20 entries

## Module 10 – Stock Visibility



## MODULE 9 – INVOICE & PAYMENT MANAGEMENT (WIREFRAME DIAGRAM)

Objective: Process supplier invoices, verify, manage payments and track payment status.



LEGEND: View, Edit, Download, Filter, More Actions

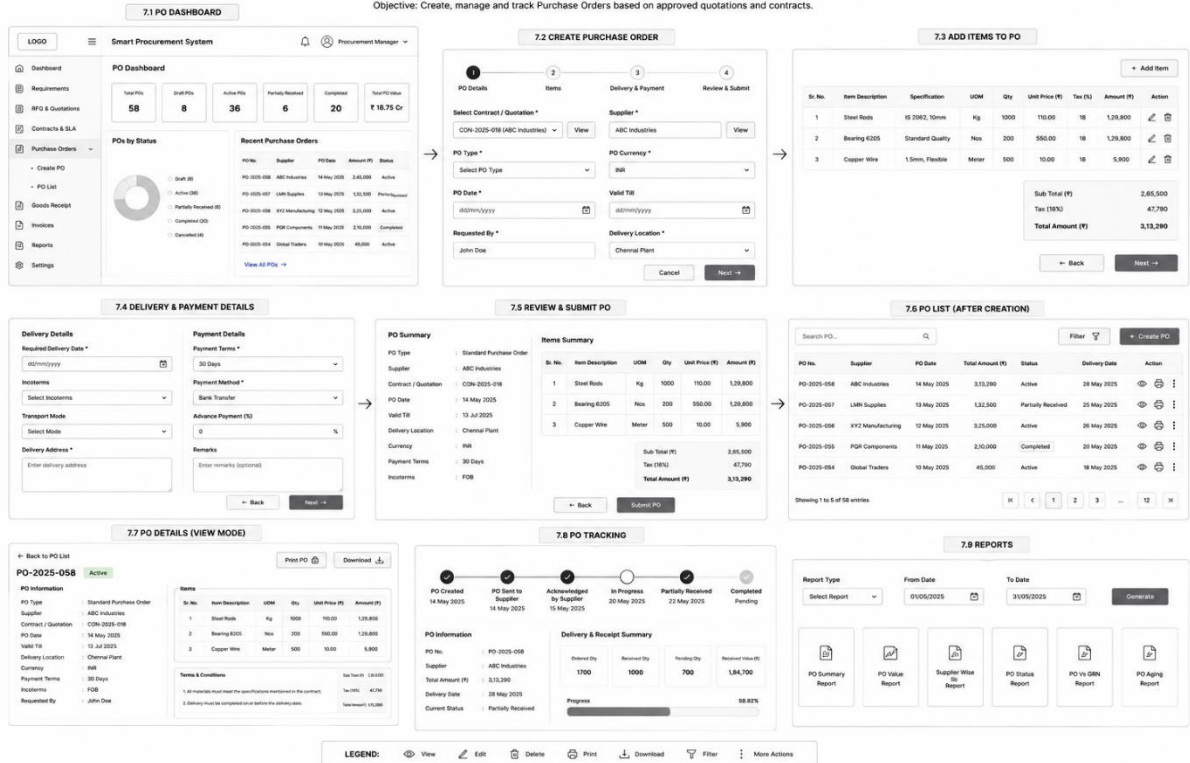
## MODULE 8 – INVENTORY & STOCK MANAGEMENT (WIREFRAME DIAGRAM)

Objective: Track stock levels, manage warehouses, material reservations and resource transfers.



## MODULE 7 – PURCHASE ORDER MANAGEMENT (WIREFRAME DIAGRAM)

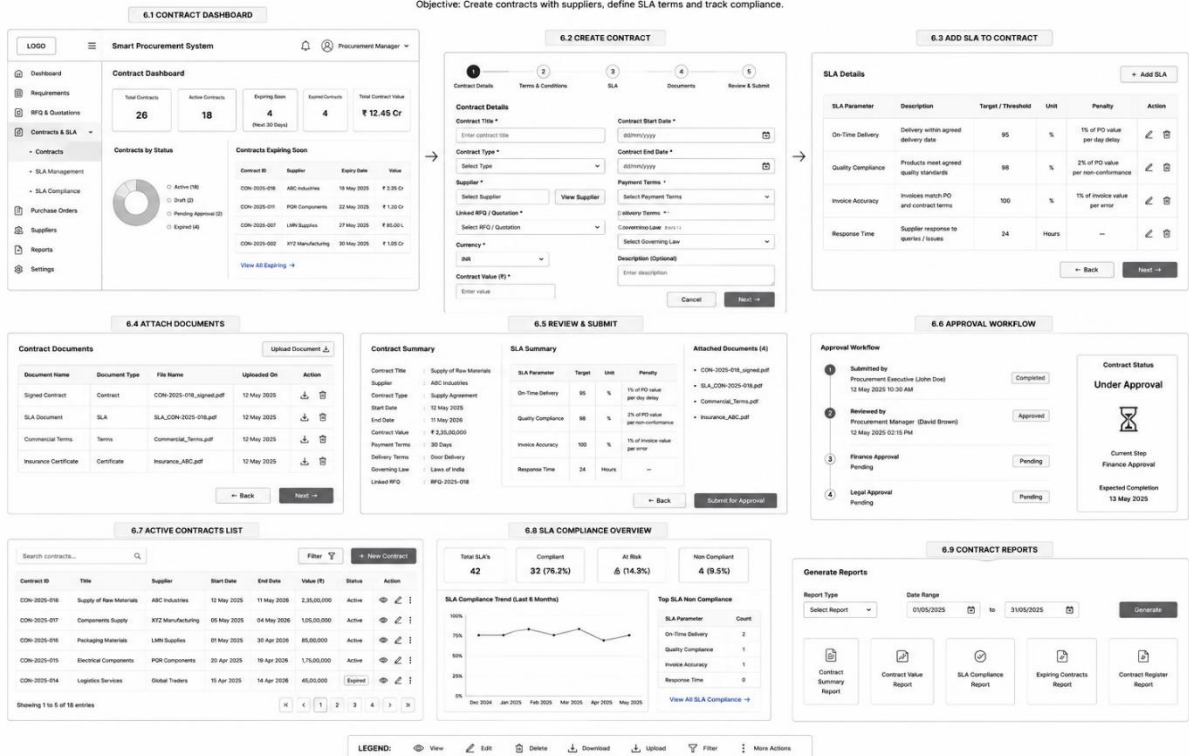
Objective: Create, manage and track Purchase Orders based on approved quotations and contracts.





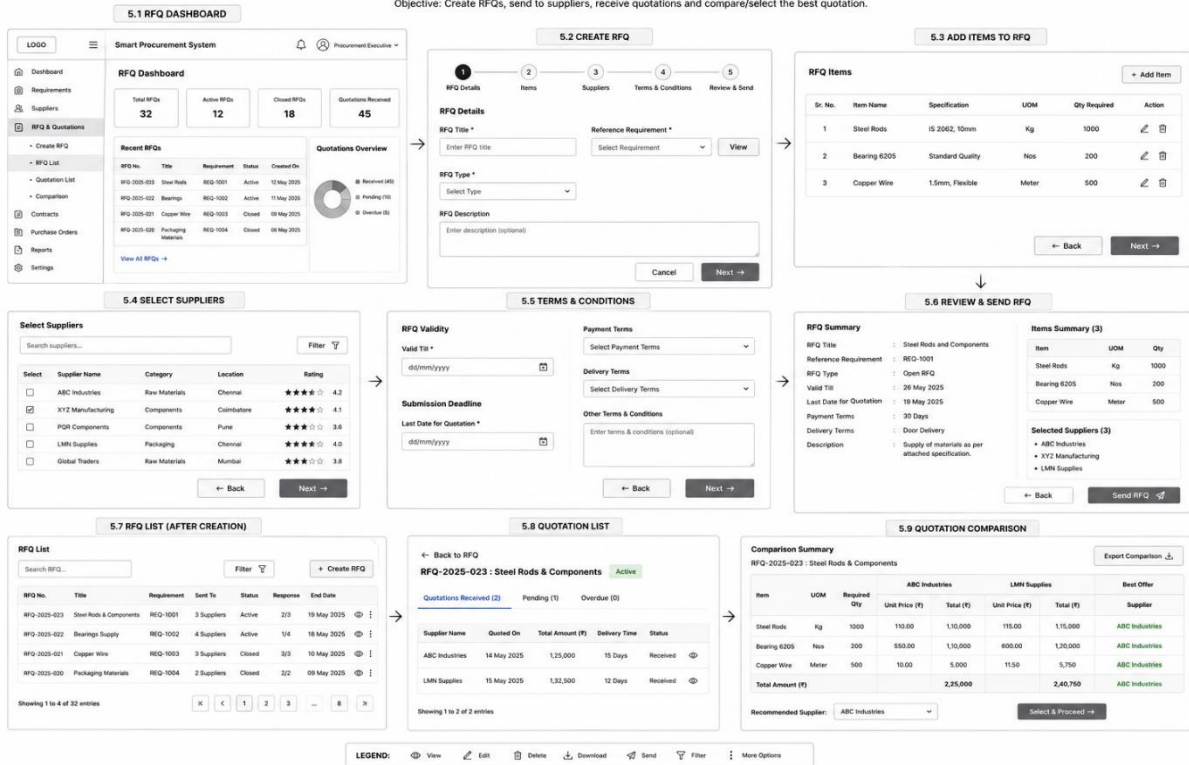
## MODULE 6 – CONTRACT & SLA MANAGEMENT (WIREFRAME DIAGRAM)

Objective: Create contracts with suppliers, define SLA terms and track compliance.



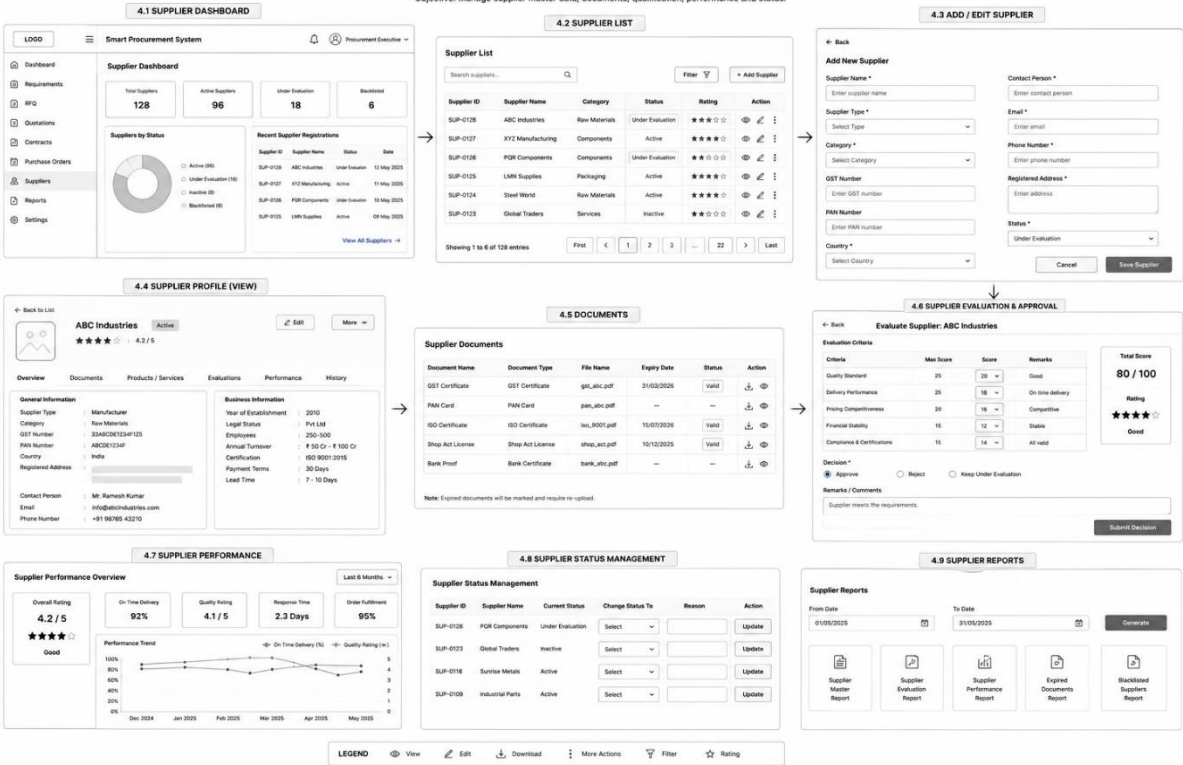
## MODULE 5 – RFQ & QUOTATION MANAGEMENT (WIREFRAME DIAGRAM)

Objective: Create RFQs, send to suppliers, receive quotations and compare/select the best quotation.



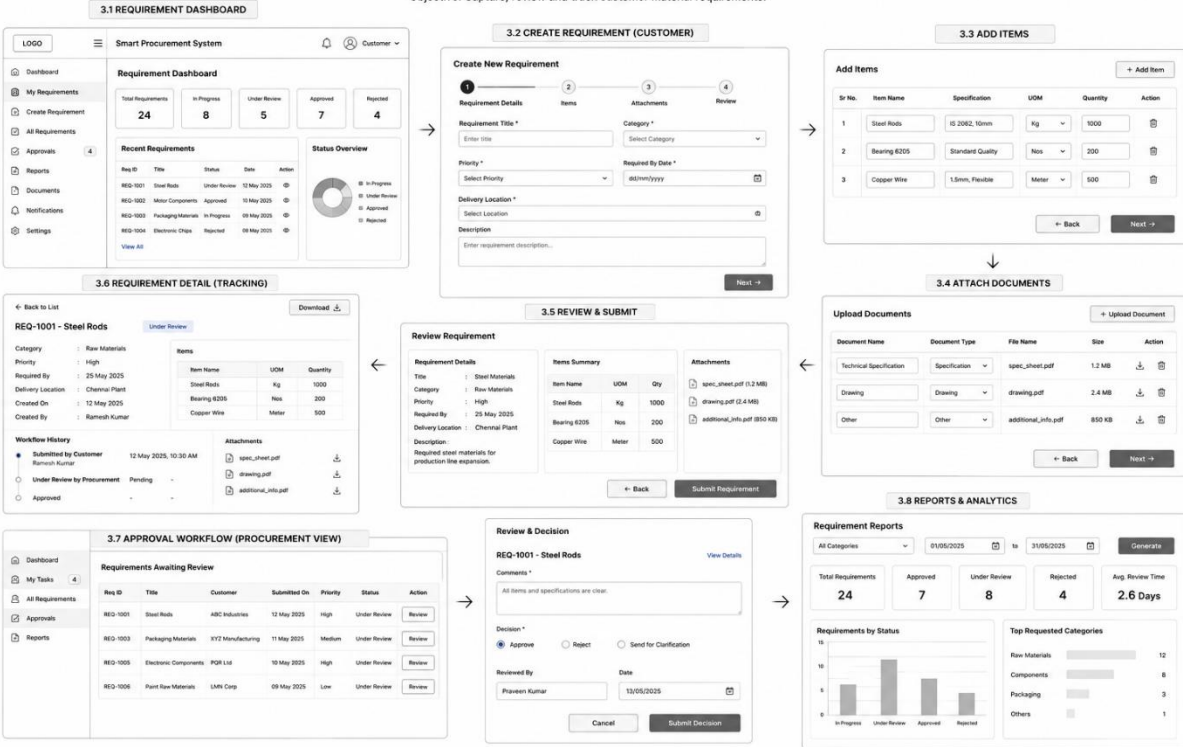
MODULE 4 – SUPPLIER MANAGEMENT (WIREFRAME DIAGRAM)

Objective: Manage supplier master data, documents, qualification, performance and status.



MODULE 3 – REQUIREMENT MANAGEMENT (WIREFRAME DIAGRAM)

Objective: Capture, review and track customer material requirements.





Objective: Admin can create employees, assign departments & roles, and manage user accounts.

## MODULE 1 – AUTHENTICATION & USER MANAGEMENT (WIREFRAME DIAGRAM)

**Objective:** Enable secure login for all users and allow Admin to manage users, roles and permissions.

This diagram illustrates the user flow for a procurement system, divided into two main sections: User Registration/Authentication and Admin Management.

### User Registration/Authentication Flow

- 1.1 LOGIN PAGE (Common for all users)**: Users log in using their email and password. The page includes a "Smart Procurement & Supplier Management System" header and a "Welcome Back!" message.
- 1.2 LOGIN PAGE (Common for all users)**: This is a duplicate of the first page, showing the login form and the "Welcome Back!" message.
- 1.3 FORGOT PASSWORD**: Users who forgot their password can click "Forgot Password?" to receive reset instructions via email.
- 1.4 RESET PASSWORD**: Users can reset their password by entering a new password and confirming it.

### Admin Management Flow

- 1.5 ADMIN DASHBOARD**: The admin dashboard shows various metrics (Total Users: 124, Employees: 58, Customers: 28, Suppliers: 38) and a list of recent user registrations. It also includes a "User Status Overview" chart.
- 1.6 USER LIST (ADMIN)**: The user list shows a table of users with columns for Name, Email, Role, Department, Status, and Action. It includes a search bar and a filter dropdown.
- 1.7 ADD / EDIT USER (ADMIN)**: The add/edit user form allows administrators to create new users or edit existing ones. It includes fields for Full Name, Email, Password, Confirm Password, Department, Role, and Status.

### Role Management

**1.8 ROLE & PERMISSION MANAGEMENT (ADMIN)**

This section includes a table for Role Management and a table for Permission Management.

| Role Name             | Description                                | Users | Action                                      |
|-----------------------|--------------------------------------------|-------|---------------------------------------------|
| Procurement Manager   | Manages procurement & sourcing activities. | 5     | <a href="#">Edit</a> <a href="#">Delete</a> |
| Procurement Executive | Handles RFQ, quotations and POs            | 12    | <a href="#">Edit</a> <a href="#">Delete</a> |
| Planning Executive    | Handles planning and inventory operations  | 8     | <a href="#">Edit</a> <a href="#">Delete</a> |
| Finance Executive     | Handles invoices, payments and finance     | 7     | <a href="#">Edit</a> <a href="#">Delete</a> |
| Auditor               | Audit activities and reports               | 4     | <a href="#">Edit</a> <a href="#">Delete</a> |

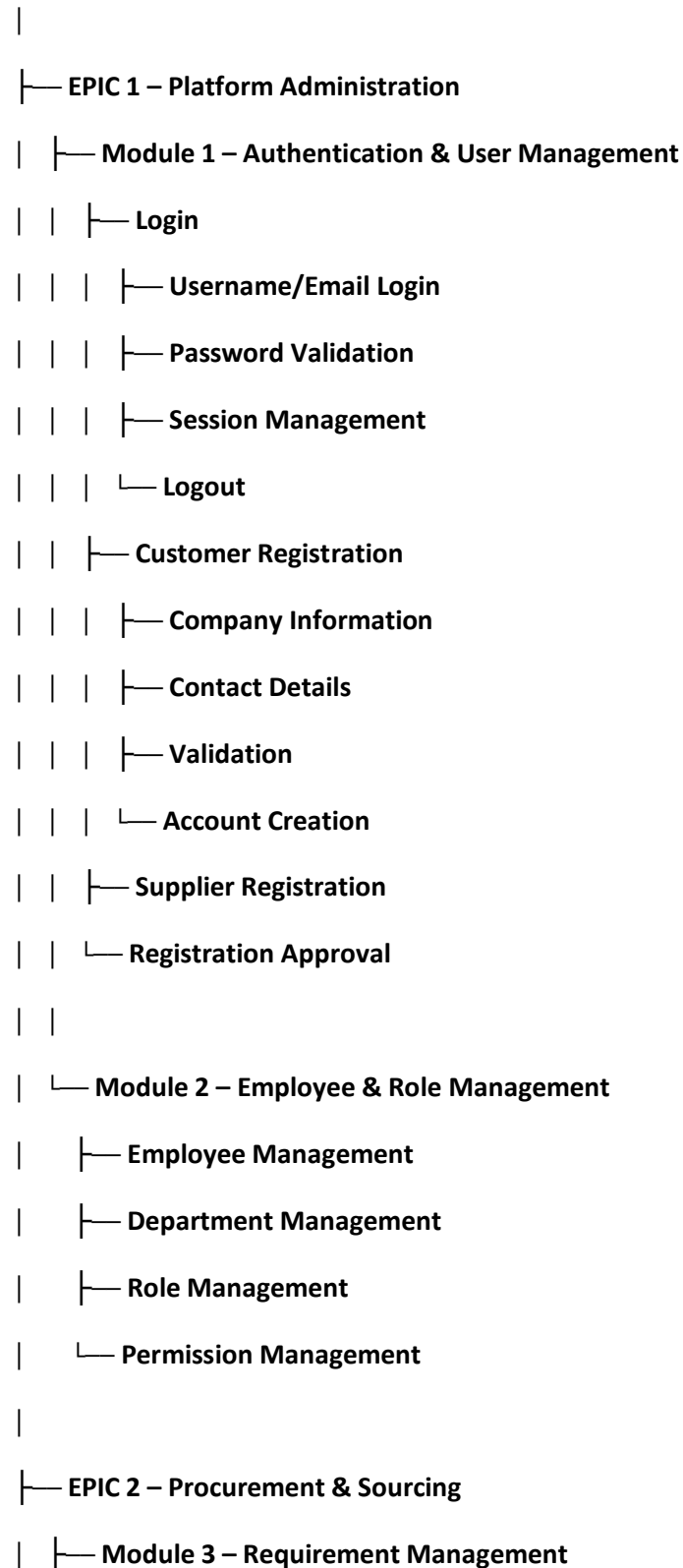
| Module / Feature | View                                | Create                   | Edit                     | Delete                   | Approve                             |
|------------------|-------------------------------------|--------------------------|--------------------------|--------------------------|-------------------------------------|
| Requirements     | <input checked="" type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/>            |
| Suppliers        | <input checked="" type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/>            |
| RFQ              | <input checked="" type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/>            |
| Quotations       | <input checked="" type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/>            |
| Purchase Orders  | <input checked="" type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input checked="" type="checkbox"/> |
| Contracts & SLA  | <input checked="" type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/>            |
| Reports          | <input checked="" type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/>            |

### Notes (Flow Summary)

- Users land on the system and login using email & password.
- Customers and Suppliers can self-register.
- Admin approves registrations (handled in other modules).
- Admin creates employees, assigns roles and departments.
- Roles have permissions which control module access.
- After login, users are redirected to their respective dashboards based on their role.

## Functional Decomposition Diagram

### Procurement & Sourcing Platform



- | | | └─ **Submit Requirement**
- | | | └─ **Review Requirement**
- | | | └─ **Requirement Tracking**
- | | └─ **Requirement History**
- | |
- | └─ **Module 4 – Supplier Management**
- | | | └─ **Supplier Profile**
- | | | └─ **Document Verification**
- | | | └─ **Performance Evaluation**
- | | └─ **Supplier Status**
- | |
- | └─ **Module 5 – RFQ & Quotation**
- | | | └─ **RFQ Creation**
- | | | └─ **RFQ Distribution**
- | | | └─ **Quotation Submission**
- | | └─ **Quotation Comparison**
- | |
- | └─ **Module 6 – Contract & SLA**
- | | | └─ **Contract Management**
- | | | └─ **SLA Configuration**
- | | | └─ **Contract Lifecycle**
- | | └─ **Contract Dashboard**
- | |
- | └─ **Module 7 – Purchase Order**
- | | └─ **PO Creation**
- | | └─ **Approval Workflow**
- | | └─ **Supplier Response**

- |   └─ PO Tracking
- |
- |   └─ EPIC 3 – Resource Planning
- |   └─ Module 8 – Material Requirement Planning
- |   |   └─ Bill of Materials
- |   |   └─ Requirement Planning
- |   |   └─ Procurement Plan
- |   |   └─ Planning Dashboard
- |   |
- |   └─ Module 9 – Purchase Requisition
- |   |   └─ Standard Requisition
- |   |   └─ Emergency Requisition
- |   |   └─ Approval
- |   |   └─ Tracking
- |   |
- |   └─ Module 10 – Stock Visibility
- |   |   └─ Warehouse Stock
- |   |   └─ Available Stock
- |   |   └─ Low Stock Monitoring
- |   |   └─ Inventory Reports
- |   |
- |   └─ Module 11 – Material Reservation & Transfer
- |   |   └─ Material Reservation
- |   |   └─ Material Allocation
- |   |   └─ Internal Transfer
- |   |   └─ Reservation Tracking
- |

- └─ **EPIC 4 – Finance & Audit**
  - └─ **Module 12 – Invoice Verification**
    - | └─ **Invoice Upload**
    - | └─ **Three-Way Matching**
    - | └─ **Invoice Validation**
    - | └─ **Approval**
    - |
  - └─ **Module 13 – Payment & SLA Compliance**
    - | └─ **Payment Processing**
    - | └─ **SLA Monitoring**
    - | └─ **Penalty Management**
    - | └─ **Payment History**
    - |
  - └─ **Module 14 – Audit & Analytics**
    - └─ **Audit Logs**
    - └─ **Procurement Analytics**
    - └─ **Supplier Performance**
    - └─ **Financial & Compliance Reports**

# 1. Customer Sidebar

The customer initiates procurement requests and tracks their progress.

CUSTOMER

- 🏠 Dashboard
- 📁 Requirements
  - └─ Create Requirement
  - └─ My Requirements
  - └─ Requirement History

 Purchase Orders

- View Purchase Orders
- Track Orders

 Contracts

- My Contracts
- SLA Details

 Reports

- Order Status
- Requirement Reports

 Profile

 Settings

 Logout

## Customer Modules

- Module 1 (Authentication)
- Module 3 (Requirement Management)
- Module 6 (View Contracts)
- Module 7 (Track Purchase Orders)

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## 2. Supplier Sidebar

The supplier receives RFQs, submits quotations, manages contracts, responds to POs, and tracks payments.

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SUPPLIER

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 Dashboard

 Company Profile

- Company Details
- Documents
- Product Categories

 RFQ Management

- Received RFQs
- Submit Quotation
- Quotation History

 Contracts

- Active Contracts
- SLA Details

 Purchase Orders

- Received POs
- Accept / Reject PO

- └─ Delivery Schedule
- Invoices
  - └─ Upload Invoice
  - └─ Invoice Status
- 💰 Payments
  - └─ Payment Status
  - └─ Payment History
- 📊 Performance
  - └─ Supplier Rating
  - └─ Delivery Performance
- 👤 Profile
- ⚙️ Settings
- 🚪 Logout

## Supplier Modules

- Module 1
- Module 4
- Module 5
- Module 6
- Module 7
- Module 12
- Module 13

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## 3. System Administrator Sidebar

The System Administrator manages the complete platform.

- 
- SYSTEM ADMIN
- 
- 🏠 Dashboard
- 
- Platform Administration
- 
- 🔒 Authentication & Users
  - 👤📁 Employees & Roles
- 
- Procurement
- 
- 📋 Requirement Management

 Supplier Management

 RFQ & Quotations

 Contract & SLA

 Purchase Orders

---

Resource Planning

---

 Inventory Management

 Purchase Requisition

 Stock Visibility

 Material Reservation

---

Finance

---

 Invoice Verification

 Payment & SLA

 Audit & Analytics

---

 Profile

 Settings

 Logout

---

## Complete Module Mapping

| Module                     | Customer | Supplier | Admin |
|----------------------------|----------|----------|-------|
| Authentication             | ✓        | ✓        | ✓     |
| Employee & Role Management | ✗        | ✗        | ✓     |
| Requirement Management     | ✓        | ✗        | ✓     |
| Supplier Management        | ✗        | ✓        | ✓     |



| Module               | Customer           | Supplier    | Admin |
|----------------------|--------------------|-------------|-------|
| RFQ & Quotation      | ✕                  | ✓           | ✓     |
| Contract & SLA       | View               | View        | Full  |
| Purchase Order       | View               | Respond     | Full  |
| Inventory Management | ✕                  | ✕           | ✓     |
| Purchase Requisition | ✕                  | ✕           | ✓     |
| Stock Visibility     | View               | ✕           | Full  |
| Material Reservation | Request (optional) | ✕           | Full  |
| Invoice Verification | ✕                  | Upload      | Full  |
| Payment & SLA        | View               | View        | Full  |
| Audit & Analytics    | Reports            | Performance | Full  |